Design of digital gloves with feedback for VR

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Abstract

This article describes the first steps in the development of a low-cost digital sensory glove that designed for use in virtual reality systems especially. Existing concepts of gloves differ in features and design, they have various functions, including feedback, tactile feedback to the electric discharge, a feeling of finger bending, finger grip strength and prediction of action and three-dimensional spatial positioning – to improve sensation and practical experience in virtual reality. Manual dynamic perception and freedom of action, common in the real world, provide instant information about objects in the virtual world. Digital gloves act not only as a remote control in VR, but also provide physical feedback for the user when they come in contact with virtual objects. This article presented an own design for inexpensive gloves that allow for proximal and distal finger joint movements, as well as position/orientation determination with an inertial measuring unit. These sensors and tactile feedback caused by the vibration patterns of the coins at the fingertips are integrated into a wireless, easy-to-use and open-source system. The design of hardware, as well as experiment plans for proof of concept, is presented.

1. Introduction

Human hands are the primary agents that physically interact with the external world, it's intricate and very important parts of the human body, which respond to different tasks of manipulations and touches, various everyday activities. The monitoring and tracking human hand motion are crucial applications in rehabilitation, skill training, including motor fine skills training, entertainment, and learning. Such tasks lead to the development of a large number of new methods of perception of the position of the hands, fingers, and approaches for the transmission of touch, for example, there an optical [1], an acoustic [2], and a magnetic sensing [3], innovative techniques such as fiber optics [4, 5], strain gauge [6, 7], Hall Effect sensing [8] methods. All the same, we can still improve these sensing methods to achieve the stringent requirements from applications in medicine and skill training, such as sensing accuracy via stability of the sensor signal and repeatability and reliability of movements, ease of wear and removal, rapid calibration, adaptation for different hand sizes, no electromagnetic interference, no temperature variation, and, as important condition, production of digital sensory glove having low cost [9].

Sensory gloves have different applications for human-machine interactions: they are used for rehabilitation [10] for studying psychological problems [11], for developing motor skills in training simulation [12–15].

2. Related works

In addition, vibrationotact feedback can improve the interaction of a person with a robot, giving the user a sense of ownership [17]. It is indisputable that the potential benefits depend on the application. For some rehabilitation robots, a combined degree of freedom may be sufficient for each finger [18], while additional ones can improve the integration of the body scheme [19] and can be crucial in manual exoskeletons [20].

Unlike commercial and rather expensive data gloves, such as CyberGlove [21], many inexpensive gloves do not provide more than one degree of freedom (DoF) per finger [19, 22]. In addition, resistive sensors [19], motion tracking based on markers [23], optical linear encoders [24] or strain sensors [25] are used to track finger movements. While the glove of [24] provides additional DoF, it lacks other important functions, such as hand grip / hand orientation. The

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glove presented in [26] provides a wireless interface for the host, but only five DoFs. Despite the fact that DJ Dochs and manual positioning / orientation are perceived, DJ Handschuh [27] is limited to one finger. Alternative approaches to motion capture are based on external tracking of the human hand by merging visual and gyroscopic data [28] or using a marker measurement [29]. Most gloves do not provide vibration feedback. One system that implements feedback and combines it with the other functions discussed above is VR Data Glove [30]. This glove is aimed at applications of virtual reality and demonstrates only one degree of freedom on each finger.

In [31] proposed a digital glove design that addresses the shortcomings of VR remote controllers in user interactivity and immersion: digital glove incorporates functions to determine wrist posture, achieve 3D positioning, sense finger curvature, sense fingertip force, limit joint movement, and provide tactile feedback using electrical shocks. The wrist posture is determined using a gyroscope that provides the yaw, pitch, and roll angles of the glove together with movement information measured using a 3D accelerometer.

In [32] presented the design of a device, termed the Wolverine system, which attaches to the tips of three fingers and the thumb. The device provides the sensation of grasping a rigid object by resisting relative motion between the fingers and thumb. It utilizes a brake-based system to provide high resistance to forces in a lightweight, low power, and low cost package.

3.1. Commercial digital gloves and its characteristics

Speaking about the development of a new design of gloves, one can not help mentioning already existing commercial analogues and showing what these solutions do not suit and why new solutions are required.

To date, the entire range of commercial gloves can be divided into two types:

1. Gloves with tactile negative feedback.
2. Gloves with power negative feedback.

The following gloves can be classified as the first type of gloves: GloveOne, CaptoGlove, Manus VR, Senso, Noitom Hi5 VR, VRfree, HaptX. This type of glove is equipped with 5 or 6 sensors with 9 degrees of freedom (9-DOF IMU) for tracking the position of the hands and fingers in space, sensors for registering the bending of fingers, as well as various means of feedback such as vibration motors, piezo sensors, electrical pulses and etc. Gloves also have capacitive areas for performing various commands, a motherboard with a microcontroller and a battery. This computer is usually passed from gloves through a wired USB connection or via Bluetooth protocols.

3.2. Disadvantages of existing commercial devices

- There is no built-in interpretation of human biosignals;
- There is no way to give feedback and perceive virtual objects as real;
- No force is used (except for Dexmo) instead of vibrotactyl feedback [33];
- Usually commercial gloves are very expensive;
- Gloves should be easy to assemble from the kit;
- Not all gloves are wireless and have low latency.

The main purpose of this design at the moment is to be able to use gloves in the telemedicine mode to restore motor functions in nervous system lesions.

4. Proposed idea

Well, we believe that ideal haptic feedback interfaces for consumer VR should be low cost, lightweight, ungrounded, while still providing force feedback that realistically simulates touching and manipulating virtual objects; that is, the interfaces should resist forces larger than finger strength at a high refresh rate with high accuracy.

Since an inexpensive glove combines finger tracking with multiple DoFs, vibro-inactivated feedback, position detection/arm orientation and wireless pairing, this article suggests a new design based on a digital glove with an open-source touchscreen and an inexpensive touch-sensitive sensor glove described in [9].

It expands the existing concept to provide these functions and uses available electronic components and simplifies development. The hardware design of the new inexpensive sensory digital glove is presented on the basis of a brief description of the preliminary glove. In addition, we will describe the planned functionality of measuring the movement of fingers and hands.

Key features of our digital glove design are the consideration of interphalangeal (IP) and metacarpophalangeal (MCP) joint flexion of the index, middle, ring, and pinky fingers as well as metacarpophalangeal I (MCP I) and carpometacarpal (CMC) joint flexion of the thumb (see Fig. 1).

In the prototype glove [9], 4.5-inch bend sensors were used on each individual finger to fix the bending and stretching. For the procedure of tracking the orientation and position of the hand in space, HTC VIVE trackers were used. For this purpose HTC VIVE trackers were fixed on the back of the glove and captured by several infrared cameras installed in the laboratory. To ensure reverse vibration tactile connection, vibration motors were attached to the tip of each finger.

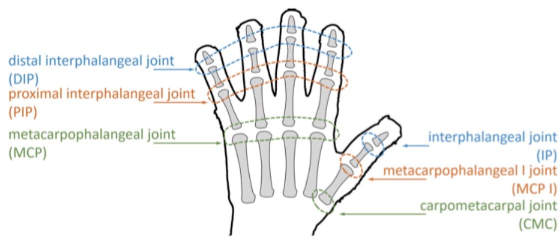

Fig. 1. Overview of the human hand articulations [9]

For the role of processing and data transfer, a microcontroller board based on Arduino was used, which in turn transmitted all the necessary values to the computer. To exchange data between the Arduino microcontroller and a computer, a USB controller is used. With further implementation, the data exchange using Bluetooth 4.0 technology will be applied.

4.1. Glove electronic implementation

The objectives of redesigning the glove leading to the creation of a new digital glove are as follows:

- feeling complex movements;
- improved positioning / orientation;
- system integration;
- advanced motion capabilities;
- improved ergonomics thanks to wireless transmission;
- detection of human biosignals for subsequent interpretation.

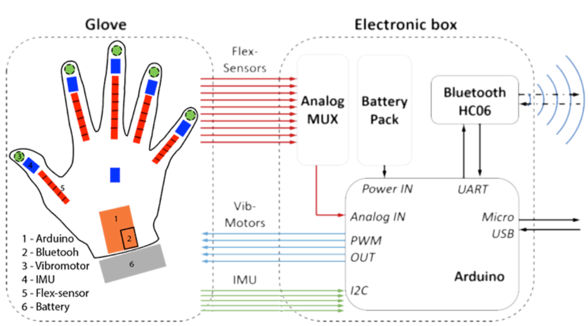

Fig. 2. Simplified block diagram of the electronic implementation

For this purpose, the new glove uses ten sensors of flexibility, visual motion tracking is replaced by an inertial measuring unit (IMU), the position of the vibration motors is optimized. The software that ensures the operation of all these sensors and functions must be completely redesigned.

As mentioned earlier, the entire tracking and monitoring system is based on the Arduino microcontroller. The choice of the Arduino board is explained by the fact that it has a relatively small size, good performance, low cost and the presence of all necessary interfaces. As shown in Figure 2, the microcontroller reads all the data from the sensor and transmits them for further processing to the computer. Also, the microcontroller receives commands from the computer to send the signal and intensity to the vibration motors, which act as tactile feedback. The connection between the microcontroller and the computer can be realized via the Micro-USB port on the microcontroller or via Bluetooth. To implement the connection via Bluetooth, an additional transceiver module (AUKRU HC-06) is installed, which is connected to the serial UART interface on ARDUINO. The IMU is connected to the microcontroller via the I2C serial interface.

The intensity of vibration feedback depends on the frequency and amplitude of the vibration signal. The frequency and amplitude are directly proportional to the input voltage of the vibration motor and can vary in the range from 50 Hz to 220 Hz.

To adjust the input voltage, a pulse width modulator (PWM) is used. The sensor glove needs five PWM outputs (vibromotors), seven serial ports (IMU and BLUETOOTH module) and six analog inputs (monitoring the battery level and 5 flex sensors). A replaceable battery pack with 2500 mAh provides enough power for a constant work of at least five hours. Figure 2 shows a block diagram of the system, its components and interfaces.

4.2. The experimental future works

As the first demonstrative concept of the design and functionality of the new digital glove, its capabilities and characteristics of its sensors should be demonstrated, as well as the advantages of separating DIP / PIP and MCP.

1. Finger Sensors: This experiment must demonstrate that the placement of the sensors is appropriate to collect valuable information about the flexion of the single phalanges.
2. To present the detection of the correct orientation by the IMU, an example of the data collected during various arm motions and poses.
3. To validate the precision and drift effects of the gyroscope, accelerometer, and magnetometer, further experiments will be performed in the future.

4. Calibration application: needed developed a calibration protocol for the sensors that capture the motion of the fingers.
5. Design of the physical limitations for the hand.
6. The finger curvature measurements are relevant to design the physical limitations for the hand.
7. Design to simulate the gripping sensation.
8. Completing the design to physically limit the hand raises the issue of how to continuously measure the finger curvature.
9. Design to measure the finger curvature: needed evaluated a previously described mechanical structural design and propose here a new design to measure the finger curvature.
10. Design of the tactile feedback by an electrical shock.
11. The current physical feedback design is based on physical limitations. Needed produce tactile feedback that is more realistic by adopting techniques used in electrical massage therapy that simulate striking, rubbing, touching, and stinging by administering electrical shocks with varying cycles.

5. Results

Presented a new sensory glove design combines the kinematic sounding of 9-DoF for all fingers, vibro-inactivated feedback, and capture of the position of the hand/orientation. These aspects are integrated into an easy-to-use and low-cost system with wireless connection to the host computer. The key components comprise five flex sensors, which are separately measuring proximal and distal finger joint motions as well as the flexion of the thumb and the thumb saddle joints. Finally, miniature vibration motors are attached to the fingertips, providing vibrotactile feedback. Despite these improvements, the overall material costs of the kit for the new digital glove is less than 300 euro.

Although a quantitative assessment of sensor data quality is missing, however earlier experiments of other researchers demonstrate that the separate sensing of proximal and distal finger joint motion can enable teleoperating grasps with increased complexity. Moreover, the additional detection of thumb saddle joint motions enables grasping of flat and soft objects without deforming them.

6. Conclusion

Future works will focus on improving the electronics and software to increase the operating frequency and latency minimization also a realization of the shown earlier experimental evaluation plan for the designed digital glove. The quality of flex sensor and IMU data should be quantitatively assessed and filter implementations for these data should be tested.

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